V.L Memorial Public School

475 Vijay Nagar

CHAPTER-10-QUESTIONS-CLASS7TH-MATHS.PDF

Class 07 - Maths (Ganita Prakash)

Section A

1. Room temperature is 32°C, lowers by 5°C every hour. After 10 hours temperature will be **[1]**

   a) 18°C
   b) -18°C
   c) 50°C
   d) -50°C

2. 4 into -2 value will be **[1]**

   a) 9
   b) -8
   c) -6
   d) 6

3. Additive inverse of -12 is **[1]**

   a) 12
   b) -12
   c) 1
   d) 0

4. Given that $(-125) \times 276 = -34500$, the value of $(-125) \times 277$ is _____. **[1]**

   a) -46325
   b) -36425
   c) -43625
   d) -34625

5. If one of the multiplier or the multiplicand is positive and the other is negative, then their product is: **[1]**

   a) positive
   b) zero
   c) neither positive nor negative
   d) negative

6. -100 divided by -4 value is **[1]**

   a) 40
   b) -25
   c) -40
   d) 25

7. $221 \times (-23 + 8) = -3315$, then the value of $-221 \times (23 - 8)$ is _____. **[1]**

   a) 3536
   b) -3315
   c) -3536
   d) 3315

8. 3 into -4 is -12, so -4 into 3 value is **[1]**

   a) 12
   b) 7
   c) 0
   d) -12

9. $(-5) + (-20) \div (-4) + 9$ is equal to _____. **[1]**

   a) -9
   b) 9
   c) -6
   d) 4

10. When multiplicand is positive, every unit decrease in multiplier makes product decrease by **[1]**

a) Multiplicand
b) Multiplier
c) 1
d) 0

11. Value of 4 into -2 is **[1]**

a) 8
b) -8
c) 2
d) -6

12. Magnitude of product does not change if we change the **[1]**

a) Numbers
b) Signs
c) Value
d) Order

13. Multiplication for integers is **[1]**

a) Associative
b) Commutative
c) Distributive
d) All of these

14. If the sum is 20 and their difference is 10, then two numbers are: **[1]**

a) 16, 4
b) 12, 8
c) 15, 5
d) 14, 6

15. The sum of two integers is -18. If one of them is -11, the other one is _____. **[1]**

a) 7
b) -29
c) -7
d) 29

16. Identify the property reflected in the following: **[1]**

$(-6) \times (-3) = (-3) \times (-6)$

a) Distributive property
b) Closure property
c) Associative property
d) Commutative property

17. The negative numbers lie to the _____ of zero. **[1]**

a) either left or right
b) right
c) left
d) neither right nor left

18. Product of a debt and a fortune will be **[1]**

a) Positive
b) Fortune
c) Zero
d) Debt

19. When multiplier is positive, we _____ tokens into the bag **[1]**

a) Break
b) Remove
c) Count
d) Place

20. -1 multiplied 4 times gives product **[1]**

a) -4
b) 4
c) -1
d) 1

21. **Assertion (A):** Multiplication of integers is distributive over addition. **[1]**

Reason (R): For any integers a, b, c, $a \times (b + c) = (a \times b) + (a \times c)$.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

22. **Assertion (A):** $24 \div (6 \div 2) = (24 \div 6) \div 2$ **[1]**

Reason (R): Associative property does not hold for the division of integers.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

23. **Assertion (A):** Zero is the additive inverse of zero. **[1]**

Reason (R): Zero is neither a positive integer nor a negative integer.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** On a number line, -6 is greater than -2. **[1]**

Reason (R): On a number line, numbers lying to the right are greater.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** $(2 \div 3) \times 4 = 2 \times (3 \div 4)$ **[1]**

Reason (R): Multiplication of integers is commutative.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** If the sum of two integers is zero, then the two integers are additive inverses of each other. **[1]**

Reason (R): Additive inverses have the same magnitude but opposite signs.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** The product of a positive integer and a negative integer is always negative. **[1]**

Reason (R): The sign of the product depends on the signs of the integers being multiplied.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. If one factor is -12 and the product is -96, find the other factor. **[1]**

29. Find the value of: -1 × (-45) **[1]**

30. Find the difference between -9 and 5. **[1]**

31. Find the value of: (-3) × 4 × (-2) **[1]**

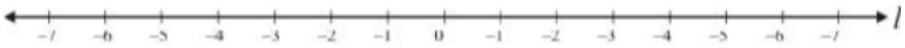
32. Find the integer which must be added to -18 to get 7. **[1]**

33. If a lift goes down 12 floors from the ground floor, what integer represents its position? **[1]**

34. If the sum of two integers is -3 and one integer is 7, find the other integer. **[1]**

Section B

35. With the help of number line find how much greater is the number 3 than -3. **[2]**



36. Given that $322 \times (-32 + 9) = -7406$, then find: **[2]**

a. $-322 \times (32 - 9)$

b. $322 \times (32 - 9)$

37. The temperature at night was -6°C. It rose by 13°C during the day. Find the temperature during the day. **[2]**

38. Write a pair of integers whose product is -24 and whose difference is 10. **[2]**

39. A shopkeeper loses ₹15 every day for 8 days. Find the total loss using integers. **[2]**

40. A bakery needs to cool down the oven after baking. Initially, the oven temperature is $180^{\circ}C$. For the first 2 hours, the temperature decreases at a rate of $5^{\circ}C$ per hour. For the next 3 hours, the temperature decreases at a rate of $4^{\circ}C$ per hour. Further the temperature decreases at the rate of $3^{\circ}C$ per hour. **[2]**

Based on the above discussion, answer the following questions:

a. What will be the oven temperature 6 hours after the cooling process begins?

b. How much total temperature has the oven lost after 6 hours?

41. An elevator descends into a mine shaft at the rate of 7m/min. If the descent starts from 5 m above the ground level, how long will it take to reach -205 m? **[2]**

42. A point moves 12 units to the left and then 20 units to the right on a number line. Find its final position. **[2]**

43. If $47 - 56 + 14 - 8 + 2 - 8 + 5 = -4$, then find the value of $-47 + 56 - 14 + 8 - 2 + 8 - 5$ without calculating the full expression. **[2]**

44. Find the value of P, if $p \times (-9) = 135$ **[2]**

45. The temperature of a cup of hot chocolate is 90°C. Every minute, the temperature drops by 4°C. What will be the temperature of the hot chocolate after 15 minutes? **[2]**

46. Replace the blank with an integer to make a true statement. **[2]**

a. $(-3) \times \quad = 27$

b. $5 \times$ _____ = (-35)

Section C

47. The temperature on a certain morning is $-\mathbf{11}^{\circ}\mathrm{C}$ at 5 a. m. If the temperature drops 3 degree at 6 a.m. and rises 5 degree at 8 a.m. and again drops 3 degree at 9 a.m. What is the temperature at 9 a.m.? **[3]**

48. A diver is at 15 m below sea level. He then descends 12 m further and later rises up by 20 m. What is his final position from sea level? **[3]**

49. A fruit merchant earns a profit of Rs. 6 per bag of orange sold and a loss of Rs. 4 per bag of grapes sold. What is the number of bags of oranges to be sold to have neither profit nor loss if the number of grapes bags are sold is 900 bags? **[3]**

50. A fruit merchant earns a profit of Rs. 6 per bag of orange sold and a loss of Rs. 4 per bag of grapes sold. Merchant sells 1800 bags of orange and 2500 bags of grapes. What is the profit or loss? **[3]**

51. A lift is at 12 m above ground. It goes down at the rate of 3 m/min for 7 minutes. What is its final position? **[3]**

52. Find a few more pairs of numbers from their sums and difference: **[3]**

a. sum = 32, difference = 8

b. sum = 12, difference = 24

c. sum = 0, difference = -10

53. An elevator starts from 30 m above ground level. It moves downward at the rate of 6 m per minute. After how many minutes will it reach the position -18 m? **[3]**

54. A number line shows a movement of +6, then -10. What is the final position if you start at 0? **[3]**

55. Using associative property, show that: $4 \times (-5 \times 2) = (4 \times -5) \times 2$. **[3]**

56. Find the values of x in the following expressions: **[3]**

a. $56 \times x = 168$

b. $(-132) \div x = 22$

c. $(-196) \div (-7) = x$

Section D

57. **Read the following text carefully and answer the questions that follow:** **[4]**

A lift in a building moves up and down from the ground floor (represented as 0).

Movement Record

Action	Floors moved
Moves up	+6
Moves down	-4
Moves up	+3
Moves down	-7

Questions:

i. If the lift is on the 6th floor and then moves down 4 floors, determine its new position using an integer operation. **(1)**

ii. Write a single integer expression representing the net effect of the first two movements. **(1)**

iii. Starting from the ground floor, find the position of the lift after the first three movements. **(2)**

OR

Find the final position of the lift after all four movements. **(2)**

58. **Read the following text carefully and answer the questions that follow:** **[4]**

A school conducted a test with the following marking scheme:

- Correct answer: +4 marks
- Incorrect answer: -2 marks
- Total questions: 30

Student Data

Student	Correct Answers	Final Score

Aman	18	48
Neha	5	-6

Questions:

a. How many marks are awarded for one correct answer? **(1)**

b. What integer represents the marks for one incorrect answer? **(1)**

c. Find the number of incorrect answers given by Aman. **(2)**

OR

Find the number of incorrect answers given by Neha. **(2)**

59. **Read the following text carefully and answer the questions that follow:** **[4]**

A hill station recorded temperature changes during a winter day.

Time Period	Change per hour (°C)	Number of hours
7 a.m. - 12 noon	-3	5
12 noon - 4 p.m.	+2	4
4 p.m. - 5 p.m.	-4	1

Initial temperature at 7 a.m. = 10°C

Questions:

a. What integer represents the temperature change per hour from 7 a.m. to 12 noon? **(1)**

b. Find the total temperature change from 12 noon to 4 p.m. **(1)**

c. Find the temperature at 12 noon. **(2)**

OR

Find the final temperature at 5 p.m. **(2)**

Section E

60. A factory earns a profit of ₹8 per white cement bag and suffers a loss of ₹5 per grey cement bag. If it sells 1,200 white bags and 800 grey bags in a month, find total profit or loss. Also explain the use of integers in representing profit/loss. **[5]**

61. Verify that $a \div (b + c)$ is not equal to $(a \div b) + (a \div c)$ for each of the following symbols of a, b and c. **[5]**

a = (-10), b = 1, c = 1

62. Ria and Khushi participated in lemon race. The race was conducted in 6 parts. In the first part, Ria won by 10 seconds. In the second part she lost by 1 minute, then won by 20 seconds in the third part and lost by 25 seconds in the fourth part, she lost by 35 seconds in the fifth part and finally won by 12 seconds in the last part. Who won the race finally? **[5]**

63. A mining elevator moves at -3 m/min (downwards). **[5]**

a. What is its position after 50 minutes if it starts at 0?

b. If it starts at 25 m above ground, what is its position after 30 minutes?

64. Verify that $a \div (b + c)$ is not equal to $(a \div b) + (a \div c)$ for each of the following symbols of a, b and c. **[5]**

a = 12, b = -4, c = 2

65. Find solutions for the following problems: **[5]**

a. Both numbers are less than 10.

The distance between the two numbers on the number line is 14.

b. Both numbers are greater than-15 and less than 5

One number is 6 greater than the other number.

c. The numbers are opposites.

The distance between the two numbers on the number line is 18.

d. One number is -35.

The distance between the two numbers on the number line is 20.

e. The first number is the opposite of -17.

The second number is less than the first number.

The distance between the two numbers on the number line is 9.

66. A cement company earns a profit of around ₹ 8 per bag of white cement that is sold and simultaneously a loss of ₹ 5 per bag of grey cement that is sold. What is the number of white cement bags that must sell to have neither a profit nor loss if the total number of grey bags sold is 6,400 bags? **[5]**

67. In the following class test, (+3) marks are given for every correct answer, (-2) marks are given for every the incorrect answer and no marks are given for not attempting any question. **[5]**

a. Radhika scored 20 marks. If she has got around 12 correct answers, then how many questions has she attempted that are incorrect?

b. Mohini scores -5 (negative) marks on this test, and though she has got seven correct answers. How many questions has she attempted incorrectly?

68. Mohan deposits Rs. 3,500 in his bank account and withdraws Rs. 1,650 from it the next day. The following day he deposits Rs. 1,200 in the morning and withdraws Rs. 1,500 in the evening. How much money is in his account if he already had Rs. 5,000 in the beginning? **[5]**

69. The temperature at 12 noon was 10 degrees C above zero. If it decreases at the rate of 2C per hour until midnight, at what time would the temperature be eight $^{\circ}C$ below zero? Also, What would be the temperature at midnight? **[5]**

Section F

70. **Fill in the blanks:** **[6]**

(a) _____ $\div 14 = -25$ **[1]**

(b) If one integer is positive and the other is negative, their product is _____. **[1]**

(c) The additive inverse of the integer -15 is _____. **[1]**

(d) If the sum of two integers is 0, then the two integers are called _____ inverses of each other. **[1]**

(e) $7\times$ _____ $= -14$ **[1]**

(f) The sum of two numbers is 15 and their difference is 7, then two numbers are _____ and _____. **[1]**